

7.1 (1)

	V_1	V_2	V_3	V_4	V_5	V_6
V_1	0	0	0	0	0	0
V_2	1	0	0	1	0	0
V_3	0	1	0	0	0	1
V_4	0	0	1	0	1	1
V_5	1	0	0	0	0	0
V_6	1	1	0	0	1	0

(2)

 $V_1 \wedge$ $V_2 \rightarrow V_1 \rightarrow V_4 \wedge$ $V_3 \rightarrow V_1 \rightarrow V_4 \wedge$ $V_4 \rightarrow V_1 \rightarrow V_5 \rightarrow V_6 \wedge$ $V_5 \rightarrow V_1 \wedge$ $V_6 \rightarrow V_1 \rightarrow V_3 \rightarrow V_5 \wedge$

(3)

 $V_1 \rightarrow V_2 \rightarrow V_5 \rightarrow V_6 \wedge$ $V_2 \rightarrow V_3 \rightarrow V_4 \wedge$ $V_3 \rightarrow V_4 \wedge$ $V_4 \rightarrow V_5 \wedge$ $V_5 \rightarrow V_6 \rightarrow V_1 \wedge$ $V_6 \rightarrow V_1 \rightarrow V_4 \wedge$

(4)

 $\{V_2, V_3, V_4, V_6\}, \{V_1\}, \{V_5\}$

7.2

1: 7, 9

6: 1, 2

2: 3, 7

7: 3, 10

3: 4, 8

8: 1, 4, 9

4: 5, 9

9: 5, 7, 10

5: 6, 10

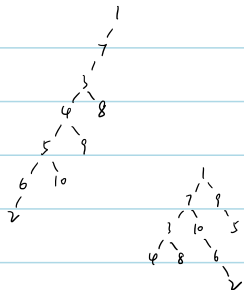
10: 1, 6

(1)

DFS: 1, 7, 3, 4, 5, 6, 2, 10, 9, 8

(2)

BFS: 1, 7, 9, 3, 10, 5, 4, 8, 6, 2



S	P[A]	P[B]	P[C]	P[D]	P[E]	P[F]	P[G]
---	------	------	------	------	------	------	------

A	✓	A	A	A	✓	✓	✓
---	---	---	---	---	---	---	---

A, C	✓	A	A	A	C	C	✓
------	---	---	---	---	---	---	---

A, C, F	✓			F	C	C	F
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A, C, F, E	✓			F			F
------------	---	--	--	---	--	--	---

A, C, F, E, D	✓						D
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A, C, F, E, D, G	✓						D
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A, C, F, E, D, G, B	✓						D
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7.3 (1)

A B C D E F G H

A	0	4	3				
B	4	0	5	9			
C	3	5	0	5			
D		5	5	0	7	6	5
E		9		7	0	3	
F				6	3	0	2
G				5	2	0	6
H				5	4	6	0

空白处为 ∞ A开始: $A \rightarrow C, A \rightarrow B, B \rightarrow D, D \rightarrow H, D \rightarrow G, G \rightarrow F, F \rightarrow E$. $W_{min} = 76$

(2)

 $A \rightarrow B \rightarrow C \rightarrow 3 \wedge$ $B \rightarrow A \rightarrow C \rightarrow 5 \rightarrow D \rightarrow 5 \rightarrow E \rightarrow 9 \wedge$ $C \rightarrow A \rightarrow 3 \rightarrow B \rightarrow 5 \rightarrow D \rightarrow 5 \rightarrow H \rightarrow 5 \wedge$ $D \rightarrow B \rightarrow 5 \rightarrow C \rightarrow 5 \rightarrow E \rightarrow 7 \rightarrow F \rightarrow 6 \rightarrow G \rightarrow 5 \rightarrow H \rightarrow 4 \wedge$ $E \rightarrow B \rightarrow 9 \rightarrow D \rightarrow 7 \rightarrow F \rightarrow 3 \wedge$ $F \rightarrow D \rightarrow 6 \rightarrow E \rightarrow 3 \rightarrow G \rightarrow 2 \wedge$ $G \rightarrow D \rightarrow 5 \rightarrow F \rightarrow 2 \rightarrow H \rightarrow 6 \wedge$ $H \rightarrow C \rightarrow 5 \rightarrow D \rightarrow 4 \rightarrow G \rightarrow 6 \wedge$

Kruskal: 2 F-G

3 A-C E-F

6 D-F G-H

4 A-B D-H

7 D-E

5 B-C B-D C-D C-H D-G

9 B-E

F-G A-C E-F A-B D-H ~~B-C~~ ~~B-D~~ ~~D-H~~ D-G $\Rightarrow$ F-G A-C E-F A-B D-H B-D D-G

7.4

 $V_1 < V_2, V_5 < V_6 < V_3 < V_4$ 3种 $V_1, V_5, V_6, V_2, V_3, V_4$ $V_5, V_1, V_6, V_2, V_3, V_4$ $V_5, V_6, V_1, V_2, V_3, V_4$

7.5

S	D[A]	D[B]	D[C]	D[D]	D[E]	D[F]	D[G]
---	------	------	------	------	------	------	------

A	0	15	2	12	∞	∞	∞
---	---	----	---	----	----------	----------	----------

A, C	0	15	2	12	10	6	∞
------	---	----	---	----	----	---	----------

A, C, F				11	10	6	16
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A, C, F, E				11			16
------------	--	--	--	----	--	--	----

A, C, F, E, D							10
---------------	--	--	--	--	--	--	----

A, C, F, E, D, G							14
------------------	--	--	--	--	--	--	----

A, C, F, E, D, G, B							14
---------------------	--	--	--	--	--	--	----

 $A \rightarrow C \quad d=2$ $A \rightarrow C \rightarrow F \rightarrow D \quad d=11$ $D = (0, 15, 2, 11, 10, 6, 14)$

A B C D E F G

 $A \rightarrow C \rightarrow F \quad d=6$ $A \rightarrow C \rightarrow F \rightarrow D \rightarrow G \quad d=14$ $A \rightarrow C \rightarrow E \quad d=6$ $A \rightarrow B \quad d=15$

```

7.8 void InDegree ( ALGraph G, int Indegree[]) {
    int i;
    ArcNode *p;
    for (i=0; i < G.vexnum; i++) Indegree[i]=0;
    for (i=0; i < G.vexnum; i++) {
        p = G.vertices[i].firstarc;
        while (p != NULL) {
            Indegree[p->adjvex]++;
            p = p->nextarc;
        }
    }
}

```

```

7.9 bool DFSPath (ALGraph G, int v, int j, bool visited[]) {
    ArcNode *p;
    visited[v] = true;
    if (v==j) return true;
    p = G.vertices[v].firstarc;
    while (p != NULL) {
        int w = p->adjvex;
        if (!visited[w]) {
            if (DFSPath (G, w, j, visited)) {
                return true;
            }
        }
        p = p->nextarc;
    }
    return false;
}

```

```

bool ExitPath (ALGraph G, int i, int j) {
    bool visited[MAX_VERTEX_NUM];
    for (int k=0; k < G.vexnum; k++)
        visited[k] = false;
    return DFSPath (G, i, j, visited);
}

```

```

7.10 bool BFSPath (ALGraph G, int i, int j) {
    bool visited[MAX_VERTEX_NUM];
    for (int k=0; k < G.vexnum; k++)
        visited[k] = false;
    int v, w;
    ArcNode *p;
    InitQueue (&Q);
    visited[i] = true;
    EnQueue (&Q, i);
    while (!QueueEmpty (Q)) {
        DeQueue (&Q, &v);
        p = G.vertices[v].firstarc;
        while (p != NULL) {
            w = p->adjvex;
            if (w==j) return true;
            if (!visited[w]) {
                visited[w] = true;
                EnQueue (&Q, w);
            }
            p = p->nextarc;
        }
    }
    return false;
}

```